



Quality Control Plan

STOCK-QP-011_B

Ss8-165-3

Revision	Date	Author	Checked By	Amendments
A	10/07/2024	R Barnbrook	A Franklin	Initial Issue
B	10/03/2026			Added straightness requirement, updated coating details

Project	STOCK	Part Number	Ss8-165-3
Specification	SPEC-ACC-Ss8 SPEC-ACC-FE6	Drawing Number	ACC-Ss8-165-3 ACC-Fe6-01

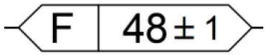
Quality Control Plan

Sample [] Trial-production [] Mass production [✓]	Main contact/Tel:	Prepared Date: 10/03/2026	Revise Date:
Core Team:		Approved date about the project (if necessary):	
		Approved date about the quality (if necessary):	
Supplier: Sudelan	Supplier No.:	Approved date (if necessary):	Approved date about others (if necessary):

No.	Description	Characteristic			Method					FAT		
		Item	Part Name	Content	Criteria/ Nominal Value / Tolerance	Relevant Standard / Document	Measuring Equipment	Min Sample Frequency	Control Method			
1	Material	1	Screwspike Ss8-165-3	Tensile Strength		≥ 500 MPa	UIC 864-1 SPEC-ACC-Ss8	Universal Testing Machine	1 per batch of raw material	Test report	Document review only	
				Yield Strength		$\sigma \geq 300\text{MPa}$						
				Elongation		$\delta \geq 20\%$						
		2	Spring Washer 38Si7	Chemical Composition Test	C	0.35-0.42%		Delkor Specification SPEC-ACC-FE6	Various	1 per batch of raw material	Test report	Document review only
					Si	1.50-1.80%						
					Mn	0.50-0.80%						
					P	≤0.025%						
					S	≤0.025%						
				Tensile Strength		1180-1370Mpa			Universal Testing Machine			
				Yield Strength		≥1030Mpa						
				Elongation at break		≥6.0%						
				Reduction of area after break		≥20%						
				Impact energy at room temperature KU ₂		≥9J						

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No.	Description	Characteristic			Method					FAT
		Item	Part Name	Content	Criteria/ Nominal Value / Tolerance	Relevant Standard / Document	Measuring Equipment	Min Sample Frequency	Control Method	
2	Finished Product	1	Fe6 Spring Washer	Springiness on Release Test	- Meet value on manufacturing drawing for relieving spring way - Overall height after test must be within limits of manufacturing drawing	PROC-SWVT	Various	Single Sampling Plan, Special Inspection Level S-I, AQL 2.5 (ISO 2859-1)	Test report	Document review only
				Texture Test	Smooth broken section without trace of internal cracking	Delkor Specification SPEC-ACC-FE6	Visual			
				Torsion Test	No cracks after twisting					
				Vickers Hardness	430 – 515 HV30					
				Dimensional accuracy	Inspection dimensions shown in reference document as per example below 	ACC-FE6-01	Various	Single Sampling Plan, General Inspection Level I, AQL 2.5 (ISO 2859-1)		
				Appearance	Identification and Markings as per reference document	ACC-FE6-01	Visual Inspection			
				Coating Thickness	Hot Dip Galvanised Minimum Average ≥ 53µm	ISO 1461	Thickness Gauge	Single Sampling Plan, Special Inspection Level S-I, AQL 2.5 (ISO 2859-1)		

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No.	Description	Characteristic			Method					FAT
		Item	Part Name	Content	Criteria/ Nominal Value / Tolerance	Relevant Standard / Document	Measuring Equipment	Min Sample Frequency	Control Method	
	Finished Product	2	Screwspike Ss8-165-3	Dimensional Accuracy	Inspection dimensions shown in reference document as per example below: 	ACC-Ss8-165-3 ACC-Fe6-01	Various	Single Sampling Plan, General Inspection Level I, AQL 2.5 (ISO 2859-1)	Test report	Document review only
				Identification	As per drawing		Visual Inspection			
				Straightness	<1mm from shank centreline	UIC864-1V	Various			
				Coating Thickness	Hot Dip Galvanised Mean ≥ 55µm	ISO 1461 / AS 1214	Thickness Gauge	Single Sampling Plan, Special Inspection Level S-1, AQL 2.5 (ISO 2859-1)		